Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3311-WE01

Title: (BEng/MEng)

Geotechnics 3

Time Allowed:	2 hours
Additional Material provided:	Yes, formula sheet included as part of exam paper
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Q1

Figure Q1 sketches an aquifer that connects a water reservoir with a river through a deep layer of gravelly sand of length 3m, thickness 0.4m and permeability $k = 10^{-3}$ m/s. The water level of the reservoir is 2m and of the river is 1.5m. The materials above and below the layer of gravels can be considered impermeable.

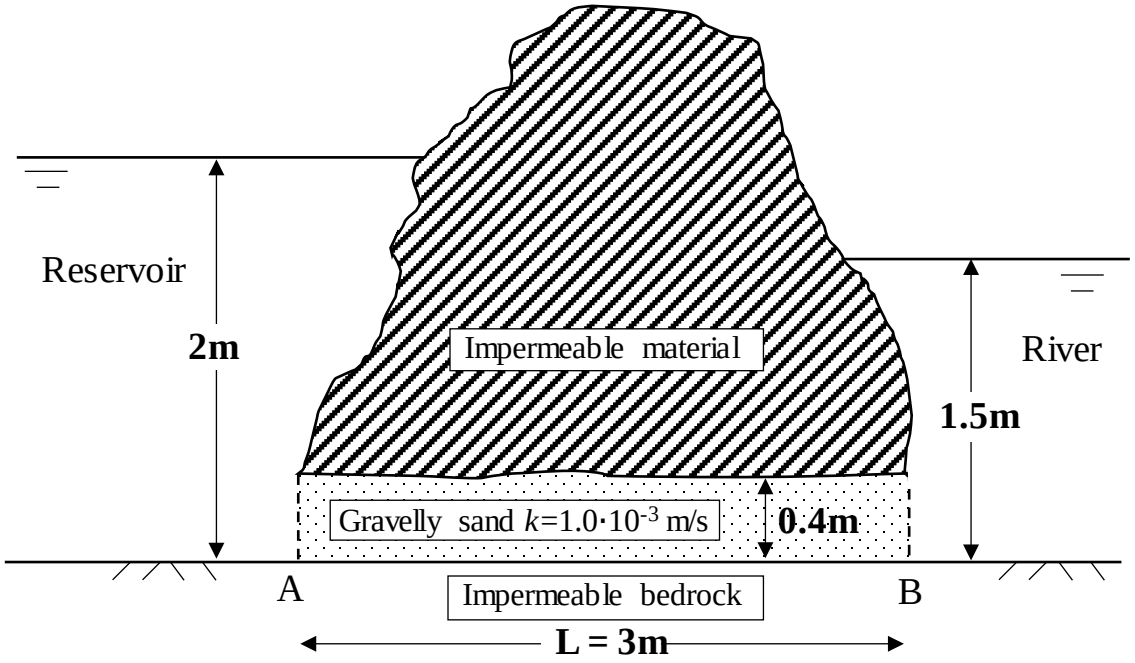


Figure Q1 (Plot to scale)

(i) Is the aquifer confined or unconfined? Justify your answer. [5%]

(ii) Discuss and identify the appropriate boundary conditions at the vertical sections A and B (see Figure Q1). [15%]

(iii) Derive the equation of the total head along AB explaining all the steps assumed with your own words and plot the total variation of total head with the horizontal direction. [15%]

(iv) Calculate the volumetric rate per day of the aquifer. [10%]

(v) In order to reduce the volumetric rate, a geotechnical engineer proposes to inject grout into the aquifer from the ground surface. Discuss with your own words the suitability/unsuitability of such idea. [5%]

(vi) Which value of permeability would be needed in the layer of gravelly sand to obtain a volumetric rate of 1m^3/day? Justify your answer. [15%]

(vii) Draw a flownet of the aquifer including all relevant boundary conditions, flowlines and equipotentials. [10%]

(viii) Calculate the volumetric flow rate using the flownet with the original value of permeability ($k = 10^{-03}$ m/s). [10%].

(ix) Compare and discuss the values for the volumetric flow rate obtained in (iv) and (viii). Is the number of flownets and equipotentials considered in (vii) important to this discussion? Justify your answers with your own words? [15%].

continued

Q2

A cylindrical sample of a normally consolidated saturated London clay is taken from a depth of 10m below the horizontal ground surface. The bulk unit weight of this saturated clay is 20kN/m^3, the friction angle $\varphi' = 22°$, the gradient of the isotropic normal compression line is $\lambda = 0.16$, the intercept of the isotropic normal compression line is $N = 2$ and the gradient of an isotropic unloading-reloading line is $\kappa = 0.05$. Assume that the water table is at the ground surface.

The cylindrical sample is placed in a triaxial apparatus and brought to the initial in-situ stress state.

(i) Estimate the initial in-situ value of vertical effective stress. [10%]
(ii) Estimate the initial in-situ value of horizontal effective stress. [15%]
(iii) Based on (i) and (ii), what is the initial value of specific volume? [15%]
(iv) What is the value of preconsolidation stress? [10%]

From this initial condition, the sample is sheared, under drained triaxial compression conditions, to critical state.

(v) Sketch, in the q:p' and v:lnp' planes, the effective and total stress paths from the in-situ initial state to critical states, indicating all relevant information. [10%]
(vi) Calculate the value of deviatoric stress at critical state. [20%]
(vii) What is the value of mean effective stress at critical state? [10%]
(viii) Calculate the value of specific volume at critical state. [10%]

continued

Q3

A retaining wall is constructed on a site to provide stability to a sand fill (see Figure Q3). The site geotechnical engineer wants to ensure that the retaining wall is stable against overturning according to Eurocode 7 (Case C). The total height of the wall is 3m, its width is 2m and thickness is 0.3m as indicated in Figure Q3. The fill material consists of sand with a saturated unit weight γ=17.5kN/m^3 and a friction angle φ'=30°. The unit weight of the light concrete in the wall is 18kN/m^3. Due to a heavy rainfall, the ground water level behind the wall is at the ground level (see Figure Q3). The wall carries a variable uniform surcharge q of 20 kPa.

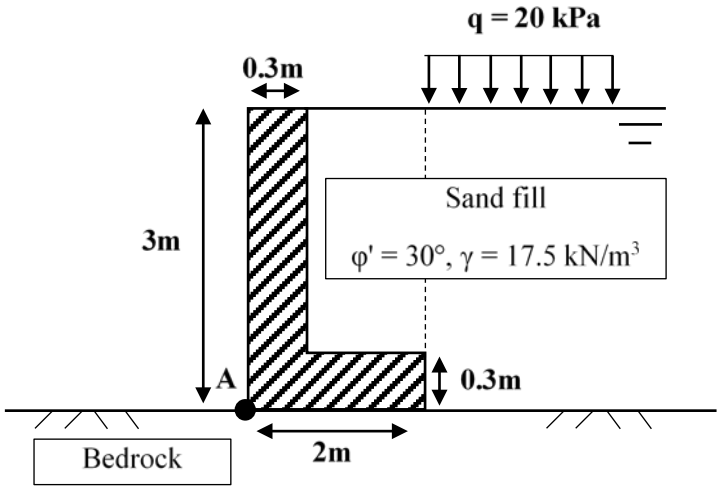


Figure Q3

(i) Assuming no friction between the sand fill and the wall, calculate the active earth pressure coefficient, K_a. [10%]
(ii) Sketch and calculate the horizontal components of pressures and corresponding thrusts acting on the wall. [10%]
(iii) Sketch and calculate the vertical components of pressures and corresponding thrusts acting on the wall. [10%]
(iv) Calculate the stability against overturning according to Eurocode 7 (Case C) by ensuring that moments about point A are stable. [20%]
(v) Assuming a friction between the sand fill and the wall of φ_w'=12°, calculate the active earth pressure coefficient, K_a. [10%]
(vi) Calculate the stability against overturning in this new situation and according to Eurocode 7 (Case C) by ensuring that moments about point A are stable. [20%]
(vii) Compare and discuss the results obtained in (iv) and (vi). [20%]

continued

Q4

A vertical cut is excavated on a site to carry out some construction work. The site geotechnical engineer wants to ensure that the vertical cut is stable (i.e. Factor of safety of at least 1.2) for an excavated height H (see Figure Q4). The material excavated is a dry silt with a unit weight γ=22kN/m^3. Major and minor principal total stresses at failure during an undrained compression triaxial test are measured as σ_1=65kPa and σ_3=10kPa, respectively.

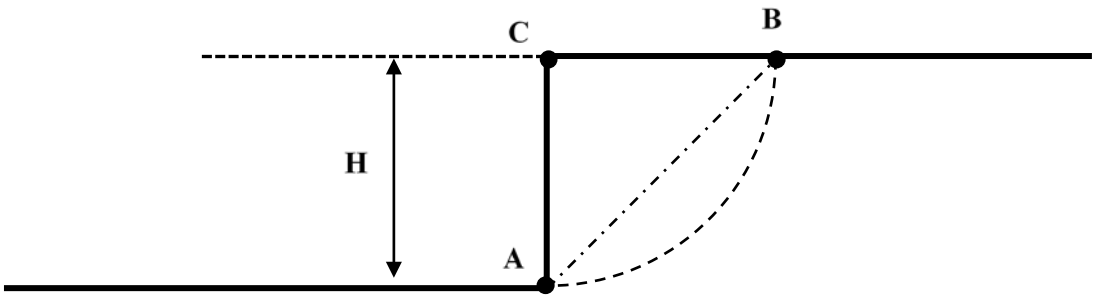


Figure Q4

Evaluate the maximum height of cutting, H, that can be achieved if the short-term failure surface can be represented by:

(i) A planar failure AB. [15%]
(ii) A circular failure AB with centre at C. [30%]
(iii) Compare and discuss the results obtained from (i) and (ii). [25%]
(iv) Calculate the critical height for (i) and (ii). Comment on the results. [30%]

continued

Phase relationships

$e=\frac{V_v}{V_s}$	$S_r=\frac{V_w}{V_v}$	$v=1+e$	$n=\frac{e}{1+e}$	$w=\frac{M_w}{M_s}$
$\rho_s=G_s\rho_w$	$wG_s=S_re$	$\rho=\frac{M_s+M_w}{V_s+V_v}$	$\rho=\frac{(G_s+S_re)\rho_w}{1+e}$	$\gamma=\rho g$
$\gamma=\frac{(G_s+S_re)\gamma_w}{1+e}$	$\gamma'=\gamma_{sat}-\gamma_w$	$\rho_d=\frac{G_s\rho_w}{1+e}$	$\rho_d=\frac{\rho}{1+w}$	$\gamma_w\approx 10$ kN/m^3

Stresses and pore pressures in soils:

$\sigma_v=\rho gz=\gamma z$	$u_w=\rho_w gz_w=\gamma_w z_w$	$\sigma_v'=\sigma_v-u$	$u_a-u_w=\frac{2T}{R}$	$h_c=\frac{2T}{\gamma_w R}$

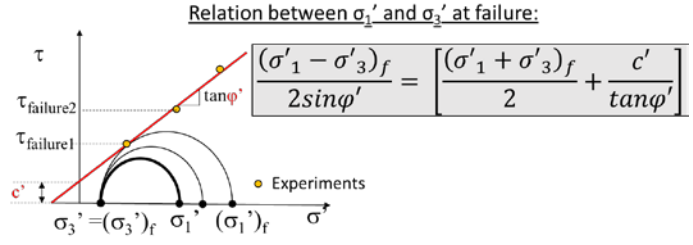
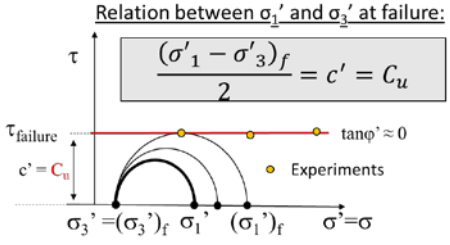
Elastic stress-strain relations:

The volumetric strain is: $\Delta\varepsilon_v=\Delta\varepsilon_1+\Delta\varepsilon_2+\Delta\varepsilon_3=\Delta\varepsilon_x+\Delta\varepsilon_y+\Delta\varepsilon_z$

$\Delta\varepsilon_x=\frac{1}{E'}\left(\Delta\sigma_x'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{xy}=\frac{\Delta\tau_{xy}}{G'}$	--
$\Delta\varepsilon_y=\frac{1}{E'}\left(\Delta\sigma_y'-\nu'\Delta\sigma_x'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{yz}=\frac{\Delta\tau_{yz}}{G'}$	$K'=\frac{E'}{3(1-2\nu')}$
$\Delta\varepsilon_z=\frac{1}{E'}\left(\Delta\sigma_z'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_x'\right)$	$\Delta\gamma_{zx}=\frac{\Delta\tau_{zx}}{G'}$	$G'=\frac{E'}{2(1+\nu')}$
$\Delta\varepsilon_v=\frac{\Delta p'}{K'}$		$\Delta\varepsilon_q^e=\frac{\Delta q}{3G'}$
$p'=\frac{\sigma'_1+\sigma'_2+\sigma'_3}{3}$		$q=\left[\frac{(\sigma_1'-\sigma_2')^2+(\sigma_2'-\sigma_3')^2+(\sigma_3'-\sigma_1')^2}{2}\right]^{1/2}$

where σ'_1, σ'_2 and σ'_3 are the principal effective stresses (i.e. $\sigma'_1\ge\sigma'_2\ge\sigma'_3$)

Failure envelopes:

$\tau=c'+\sigma'\tan\varphi'$ (Mohr-Coulomb)	$\tau=C_u$ (Tresca)
Relation between σ_1' and σ_3' at failure: 	Relation between σ_1' and σ_3' at failure: 

Ground water flow:

$h=z+\frac{u}{\rho g}=z+\frac{u}{\gamma_w}$	$q_x=Aki_x=Ak\left(-\frac{\partial h}{\partial x}\right)$	$v_x=ki_x$	$v_{true,x}=v_x\frac{1}{n}$

Constant head permeameter:	Falling head permeameter:	Pumping test:
$q=Ak\frac{h_1-h_2}{L}$	$\ln\left(\frac{h_0}{h}\right)=\frac{Ak}{aL}t$	$\left(H^2-h^2\right)=\frac{q}{\pi k}\left(\ln R_0-\ln r\right)$

Laplace's Equation (3D)	Laplace's Equation (2D, flownets)	
$\frac{\partial^2 h}{\partial x^2}+\frac{\partial^2 h}{\partial y^2}+\frac{\partial^2 h}{\partial z^2}=0$	$\delta h=\frac{\Delta h}{N_d}$	$q=k\frac{N_f}{N_d}\Delta h;\ \ u=\gamma_w\left(h-z\right)$

continued

Critical hydraulic gradient:	$i_c = \dfrac{\gamma - \gamma_w}{\gamma_w}$

1D flow through an unconfined aquifer:	1D flow through a confined aquifer:
$q_x = \dfrac{k\left(H_1^2 - H_2^2\right)}{2L}$	$q_x = Aki_x = -Ak\dfrac{\partial h}{\partial x} = -k\dfrac{H_2 - H_1}{L}D$
$h = \sqrt{\dfrac{\left(H_2^2 - H_1^2\right)}{L}x + H_1^2}$	$h = \dfrac{H_2 - H_1}{L}x + H_1$

1D straining (oedometric test conditions):

1D normal compression line:	1D swelling line:
$e = e_1 - C_c \log_{10} \sigma'_v$	$e = e_2 - C_s \log_{10} \sigma'_v$

$OCR_{1D} = \dfrac{max.\ past\ value\ of\ \sigma'_v}{current\ value\ of\ \sigma'_v} = \dfrac{\sigma'_{vc}}{\sigma'_v}$	$\sigma'_h = K_0\sigma'_v$	$K_{0nc} \approx 1 - \sin\varphi'$	$K_0 \approx (1 - \sin\varphi')\sqrt{OCR}$

Settlement of a very wide foundation:		
$\dfrac{\Delta H}{H_0} = \dfrac{e_0 - e_1}{1 + e_0}$	$\Delta H = \dfrac{H_0}{1 + e_0}\left[C_s \log_{10}\dfrac{\sigma'_{vc}}{\sigma'_{v0}} + C_c \log_{10}\dfrac{\sigma'_{v1}}{\sigma'_{vc}}\right]$	$\Delta\varepsilon_v = m_v\Delta\sigma'_v$

1D consolidation:		
$\dfrac{\partial u_e}{\partial t} = c_v\left(\dfrac{\partial^2 u_e}{\partial z^2}\right)$	$c_v = \dfrac{k}{m_v\gamma_w}$	$T_v = \dfrac{c_v t}{d^2}$

Axisymmetric conditions (triaxial test conditions):

$\Delta\varepsilon_v = \Delta\varepsilon_a + 2\Delta\varepsilon_r$	$\Delta\varepsilon_q = \dfrac{2}{3}(\Delta\varepsilon_a - \Delta\varepsilon_r)$
$p' = \dfrac{1}{3}(\sigma_a' + 2\sigma_r')$	$q = \sigma_a - \sigma_r$

where σ'_a and σ'_r are the principal effective axial and radial stresses respectively

Iso-normal compression line:	Iso-swelling line:
$v = N - \lambda \ln p'$	$v = v_\kappa - \kappa \ln p'$

Critical states:	
$q = Mp'$	$v = \Gamma - \lambda \ln p'$

where $\Gamma = N - (\lambda - \kappa)\ln(2)$

Relation between friction angle and critical state gradient:	$M = \dfrac{q}{p'} = \dfrac{6\sin\varphi'}{3 - \sin\varphi'}$

Overconsolidation ratio (OCR in terms of p')	$OCR = \dfrac{max.\ past\ value\ of\ p'}{current\ value\ of\ p'} = \dfrac{p'_0}{p'}$

Partial factors Eurocode 7 & factor of safety: (Bond & Harris, 2008; *Decoding EC7*)

$q_{Ed} \le q_{Rd}$ where, q_{Ed} is the design pressure and q_{Rd} is the design resistance	CASE	ACTIONS			Soil properties	
		Permanent		Variable	tan φ', c'	C_u
		Unfavourable	Favourable	Unfavourable		
	C	1.00	1.00	1.30	1.25	1.4

continued

Factor of safety, FoS:	$FoS = \frac{\text{available shear force, } T_f}{\text{allowable working load, } T}$; $FoS = \frac{\text{available bearing capcity, } \tau_f}{\text{allowable bearing capacity, } \tau}$

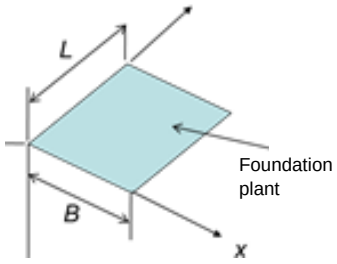
Shallow foundation settlement, δ:

Cohesionless soils:

Burland & Burbage:

$$\delta\ (mm) = f_s \cdot f_l \cdot f_t \cdot \left[\left(q'_n - \frac{2}{3}p'_0\right) \cdot B^{0.7} \cdot I_c\right]$$

where,



$f_s = \left(\frac{1.25 \cdot L/B}{\frac{L}{B} + 0.25}\right)^2$	$f_l = \frac{H}{z_l}\left(2 - \frac{H}{z_l}\right)$	$f_t = 1 + R_3 + R \cdot log\left(\frac{t}{3}\right)$

Cohesive soils: $\delta = \delta_i + \delta_c$ where,

$\delta_i\ (mm) = \mu_1 \cdot \mu_0 \cdot q_n \frac{B}{E_u}$	$\delta_c\ (mm) = \mu_g \cdot \delta_{oed}$ where, $\delta_{oed}\ (mm) = m_v \cdot \sigma'_v \cdot H$

Bearing capacity:

Brinch-Hansen (drained all soils)

$$q_f = c' \cdot N_c \cdot s_c + p'_{ov} \cdot N_q \cdot s_q + \frac{1}{2} \cdot \gamma' \cdot B \cdot N_y \cdot s_y$$

where,

$N_q = e^{\pi \tan \varphi'} \cdot \left(\tan(45 + \varphi'/2)\right)^2$	$s_q = 1 + (B'/L') \cdot \sin \varphi'$ (rectangular); $s_q = 1 + \sin \varphi'$ (circular and square)
$N_c = \left(N_q - 1\right) \cdot \cot \varphi'$	$s_c = \left(s_q \cdot N_q - 1\right)/\left(N_q - 1\right)$ (all)
$N_y = 2 \cdot \left(N_q - 1\right) \cdot \tan \varphi'$	$s_y = 1 - 0.3 \cdot (B'/L')$ (rectangular; $s_y = 0.7$ (circular and square)

depth factors are not permitted within EC7; load inclination, base and ground factors ignored.

Fine-grained soils (undrained):

$q_f = 5.14 \cdot C_u \cdot s'_c$ where,

$s'_c = 1 + 0.2 \cdot \left(\frac{B'}{L'}\right)$ (rectangular shape; for circular and square shape B'=L')

load inclination, base and ground factors are ignored.

Bearing capacity in piles:

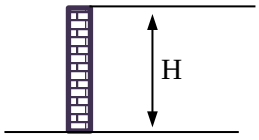
$$Q_u = Q_s + Q_b - W_p$$

Cohesionless soils:	Cohesive soils:
$Q_s = q_s \cdot A_s = A_s \cdot k_s \cdot \sigma_{v0} \cdot \tan \delta$	$Q_s = q_s \cdot A_s = \alpha \cdot C_u \cdot A_s$
$Q_b = q_b \cdot A_b = N_q \cdot \sigma'_{v0} \cdot A_b$	$Q_b = q_b \cdot A_b = N_c \cdot C_{ub} \cdot A_b$

Passive and active pressures:

$\sigma'_h = K_a \cdot \sigma'_v$ (active)	$\sigma'_h = K_b \cdot \sigma'_v$ (passive)

RANKINE'S theory: H high of the layer of soil behind the wall retaining wall, φ' is the friction angle and c' is the cohesion



In the following, $P_a = P_a' + U$ and $P_p = P_p' + U$,

where U is the resultant of the pore water pressure.

$K_a = \tan^2\left(\frac{\pi}{4} - \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1+\sin\varphi'}\right)^2 = \frac{1-\sin\varphi'}{1+\sin\varphi'}$	$K_p = \tan^2\left(\frac{\pi}{4} + \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1-\sin\varphi'}\right)^2 = \frac{1+\sin\varphi'}{1-\sin\varphi'}$

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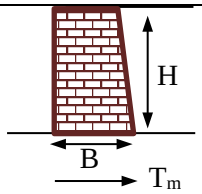
Cohesionless soils:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u$	$\sigma_h' = \sigma_v' K_a - 2c'\sqrt{K_a}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{\gamma}{2}\left(H - \frac{2C_u}{\gamma} \right)^2$	$P_a' = \frac{K_a \gamma'}{2}\left(H - \frac{2c'}{\gamma'\sqrt{K_a}} \right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = 2C_u + \sigma_v$	$\sigma_h' = \sigma_v' K_p + 2c'\sqrt{K_p}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}\left(\gamma H + 4C_u \right)$	$P_p' = \frac{1}{2} K_p \gamma' H^2 + 2c'\sqrt{K_p} H$

COULOMB'S theory: Friction and adhesion between soil and wall are φ_w and c_w, resp.

$$K_a = \left(\frac{\cos\varphi'}{1+\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}} \right)^2 \frac{1}{\cos\varphi_w'} \qquad K_p = \left(\frac{\cos\varphi'}{1-\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'}} \right)^2 \frac{1}{\cos\varphi_w'}$$

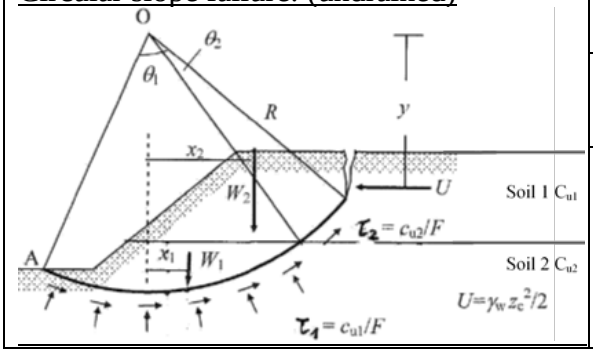
Cohesionless:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u\sqrt{1+\frac{C_w}{C_u}}$	$\sigma_h' = K_{a1}\sigma_v' - c' K_{a2}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{\gamma}{2}\left(H - \frac{2C_u}{\gamma}\sqrt{1+\frac{C_w}{C_u}} \right)^2$	$P_a' = \frac{K_{a1}\gamma'}{2}\left(H - \frac{c' K_{a2}}{\gamma' K_{a1}} \right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = \gamma z + 2C_u\sqrt{1+\frac{C_w}{C_u}}$	$\sigma_h' = \gamma z K_{p1} + c' K_{p2}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}\left(\gamma H + 4C_u\sqrt{1+\frac{C_w}{C_u}} \right)$	$P_p' = \frac{1}{2} K_{p1} \gamma' H^2 + c' K_{p2} H$

where,

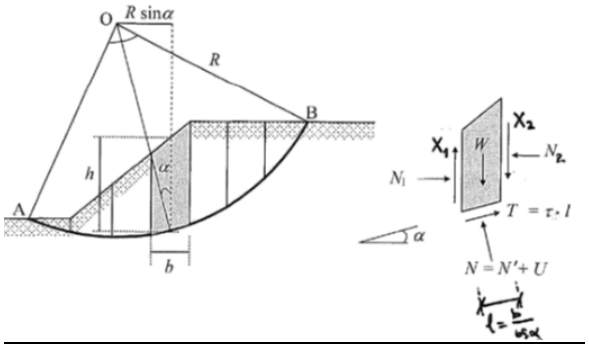
$K_{a1} = K_a$	$K_{p1} = K_p$
$K_{a2} = \frac{\cos\varphi'\left(2\sqrt{1+c_w'/c'} + \left(2+c_w'/c'\right)\sin\varphi' \right)}{\left(1+\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'} \right)^2 \cos\varphi_w'}$	$K_{p2} = \frac{\cos\varphi'\left(2\sqrt{1+c_w'/c'} - \left(2+c_w'/c'\right)\sin\varphi' \right)}{\left(1-\sin\varphi'\sqrt{1+\tan\varphi_w'/\tan\varphi'} \right)^2 \cos\varphi_w'}$
Cantilever wall in sand:	$FoS = \frac{N'\tan\varphi_w'}{T_m} = \frac{(W-U)\tan\varphi_w'}{T_m}$
Gravity wall in sand:	$FoS = \frac{N\tan\varphi_w'}{T_m}$
Gravity wall in clay: 	(Undrained) $FoS = \frac{C_w B}{T_m}$ / (Drained) $FoS = \frac{N'\tan\varphi_w' + c_w' B}{T_m}$

continued

Slope stability:

Infinite slope with parallel flow: H: layer thickness, m·H: position of water table from bedrock, β: slope angle; γ_1 (dry soil), γ_2 (saturated soil).	$FoS = \dfrac{c' + H\left(\gamma_1(1-m) + m(\gamma_2 - \gamma_w)\right)\cos^2\beta\tan\varphi'}{H\left((1-m)\gamma_1 + m\gamma_2\right)\cos\beta\sin\beta}$
Vertical cut (dry soil):	$FoS = \dfrac{4C_u}{H\gamma\sin 2\alpha}$ (undrained)
	$FoS = \dfrac{4(c' + \sigma'\tan\varphi')}{H\gamma\sin 2\alpha}$ (drained)
Circular slope failure: (undrained) 	$FoS = \dfrac{R^2\theta C_u}{Wx}$
	$FoS = \dfrac{R^2\theta C_u}{Wx + Uy}$ (with cracks)
	$FoS = \dfrac{(\theta_1 C_{u1} + \theta_2 C_{u2})R^2}{W_1 x_1 + W_2 x_2 + Uy}$ (layered soil)

$\sin 2\alpha = 2\sin\alpha\cos\alpha$, $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$ and arc-length: $\overset{\frown}{AB} = R\theta$ (θ is in radians).

Circular: (drained): 	$FoS = \dfrac{\sum_{i=1}^{n} l\left(c' + \sigma'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} = \dfrac{\sum_{i=1}^{n}\left(lc' + N'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$
Modified Bishop's b: width of the slice, l: length of the slice, α: inclination of the slice to the horizontal direction.	$FoS = \dfrac{\sum_{i=1}^{n}\left(bc' + \left(W - U\cos\alpha\right)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} M(\alpha)$ where, $M(\alpha) = \dfrac{1}{\cos\alpha\left[1 + \dfrac{\tan\alpha\tan\varphi'}{FoS}\right]}$
Fellenius:	$FoS = \dfrac{\sum_{i=1}^{n}\left(lc' + \left(W\cos\alpha - ul\right)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$

END